

How high-school pressure and autonomy support are linked to dual AI learning pathways: A cross-contextual SEM analysis

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ABSTRACT

Generative AI tools are rapidly becoming embedded in students' everyday learning, yet students' AI use varies from constructive support to compulsive dependence. Drawing on Self-Determination Theory and the Hook model, this study proposes a dual-pathway framework examining how retrospective reports of high-school learning climate relate to university students' GenAI-use patterns and perceived AI-supported learning quality through motivational and behavioral relations. Using cross-sectional survey data from university students with different high-school backgrounds and current university contexts (overall N = 508; China-HS group n = 305 and non-China-HS group n = 203; China-Uni n = 193 and Thailand-Uni n = 315), we tested the model using structural equation modeling and multi-group analyses. Results were consistent with a pressure-compulsive pathway: high-school pressure was associated with higher controlled motivation, which was in turn associated with higher compulsive AI use, lower well-being, and lower perceived AI-supported learning quality. In parallel, results supported an autonomy-adaptive pathway: high-school autonomy support and current university autonomy support were associated with higher autonomous motivation and planned trigger readiness, which were associated with more adaptive AI use, higher perceived academic performance and well-being, and higher perceived AI-supported learning quality. The model explained substantial variance in key endogenous outcomes. Scalar-level measurement invariance supported meaningful cross-group comparisons, and multi-group analyses suggested that the overall pattern of associations was broadly similar across the sampled high-school and university groupings, with differences concentrated in specific links. Overall, the findings suggest that motivational climate may be an important correlate of whether GenAI is incorporated into learning in more adaptive versus more reliance-oriented ways within the China, Thailand-linked contexts examined in this study.

1. Introduction

1.1. Broad background

Students today are undergoing an unprecedented transformation in the way they learn, as generative artificial intelligence (GenAI) rapidly reshapes educational experiences. Tools like ChatGPT reached 100 million users within two months of launch, signaling both massive interest and widespread adoption (Dwivedi et al., 2023). For many students, such tools provide empowering support, 24/7 access to explanations, feedback, and learning personalization. Yet for others, AI can become a source of anxiety or dependency, especially when used to cope with academic pressure or cognitive overload (Zawacki-Richter et al., 2019). At the same time, research on technology-supported

learning, including AI-related learning and adjacent digital-learning domains, has expanded across diverse higher-education settings, including Western contexts (Samala et al., 2025). However, existing evidence remains uneven in scope. Much of the current literature focuses on benefits, risks, or implementation challenges within specific settings, whereas less is known about whether the motivational pathways underlying students' adaptive versus reliance-oriented GenAI use operate similarly across educational systems. It is therefore premature to assume that patterns observed in one context generalize straightforwardly to others.

This mixed picture is increasingly reflected in the scholarly literature. Generative AI is increasingly described as a double-edged sword. Evidence from high-school settings in Vietnam also suggests that autonomy can be fragile in AI-supported learning: in a case study from Thai

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Nguyen Province, both students and teachers rated “monitoring and evaluating AI-assisted learning outcomes” as the weakest, underscoring the need to examine autonomy-linked pathways of later AI engagement (Anh et al., 2025). On one hand, it can foster creativity, automate feedback, and enhance personalization (Chan & Hu, 2023; Giannakos et al., 2025; Shahzad et al., 2025). On the other hand, excessive or unreflective use may be associated with weaker communication skills, diminished deep learning, academic dishonesty, and a decline in motivation (Shahzad et al., 2025). Consistent with broader evidence that problematic digital reliance can coincide with weaker learning processes, Meng et al. (2025) reported that mobile phone addiction among Chinese college students was associated with lower learning engagement. Taken together, these studies make it useful to consider whether students' AI use may reflect two broad patterns: one more adaptive and regulation-oriented, and another more reliance-oriented and avoidant. In this study, we further propose that these divergent patterns are shaped not only by the technology itself, but also by the learner's motivational profile, which may be associated with prior educational experiences. Whether students approach AI as a cognitive partner or an escape tool may be associated with whether their high-school experiences fostered autonomy and curiosity or emphasized pressure and compliance.

1.2. Theoretical grounding

To understand how prior learning environments are associated with students' present-day AI behaviors, we draw on two complementary theoretical frameworks: Self-Determination Theory (SDT) and habit-formation models from persuasive technology research.

SDT (Ryan & Deci, 2020) posits that human motivation varies along a continuum from autonomous motivation, where actions are driven by intrinsic interest or personal values, to controlled motivation, where behavior is regulated by external pressures such as rewards, punishments, or guilt. The theory emphasizes the importance of three basic psychological needs: autonomy, competence, and relatedness. When these needs are satisfied through autonomy-supportive environments such as choice-rich classrooms or supportive teacher feedback, students are more likely to engage deeply and persist in learning. Conversely, controlling learning environments, such as those marked by rigid demands, surveillance, or fear of failure, are often associated with compliance without engagement, or worse, anxiety and disengagement.

Empirical studies have confirmed these patterns. For example, Johansen et al. (2023) found that perceived teacher autonomy support positively predicted university students' autonomous motivation and academic engagement, while autonomy thwarting (i.e., pressure) was associated with higher controlled motivation and lower vitality. Similarly, Wu and Dong (2024) demonstrated that students in autonomy-supportive classrooms reported stronger psychological need satisfaction and higher subjective well-being. In parallel, research on digital behavior design shows that technologies can amplify existing motivational patterns. One widely cited model is the Hook model, which outlines a four-stage cycle: trigger → action → reward → investment (Eyal, 2014; Lukyanchikova et al., 2023). This model was initially developed to explain how digital platforms build habits, and has recently been applied to education and persuasive learning environments. In this framework, students with autonomous motivation may interact with AI tools in planned, reflective ways (e.g., asking for clarification or iterating writing), while students with controlled motivation may respond impulsively to stress-triggered cues, using AI to avoid effort or mask anxiety. In this sense, motivation functions as a psychological filter, shaping not only whether students use AI, but how they use it, and whether the outcomes are adaptive or compulsive.

1.2.1. Gap in the literature

Despite growing interest in generative AI, much of the existing research has focused on proximal predictors of students' AI use, such as

self-efficacy, academic stress, or personality traits, while largely neglecting distal developmental influences, particularly those rooted in earlier educational experiences. For instance, Hu et al. (2025) reported that among Chinese university students, compulsive ChatGPT use was associated with neuroticism and extraversion, while Zhang et al. (2024) found that academic stress and low self-efficacy significantly predicted problematic AI dependence. Although valuable, these studies emphasize present-state explanations and do not trace how students' longer-term motivational profiles, associated with high-school learning environments, might predispose them to adaptive or maladaptive AI use patterns in university.

Moreover, few studies have investigated dual-pathway models of AI behavior, where healthy (adaptive) and unhealthy (compulsive) usage styles are conceptualized in parallel. Most AI-in-education literature focuses on either positive outcomes (e.g., academic performance, engagement) or risk factors (e.g., academic dishonesty, digital addiction), but seldom both within a unified framework. This limits our understanding of how similar technologies can yield divergent behavioral outcomes depending on the learner. The integration of motivational theory (SDT) with habit-based models, such as the Hook cycle, remains rare in educational psychology. While the Hook model has been applied in software engineering and user experience design (Lukyanchikova et al., 2023), its explanatory power in the context of student-AI interaction has not been fully explored. Without this integration, we may overlook how AI platform features interact with students' motivational vulnerabilities in ways that are associated with either more self-regulated or more dependency-oriented use.

Finally, there remains a limited cross-cultural evidence base for evaluating whether these motivational pathways operate similarly across educational systems. Although some studies from East and Southeast Asian contexts suggest that pressure-intensive and autonomy-supportive learning climates may matter for students' AI engagement, it remains unclear whether comparable pathway structures would be observed in Western higher-education settings or other institutional environments. Accordingly, the present study should be understood as testing a theory-informed model in China, Thailand-linked contexts rather than as validating a universally generalizable psychological mechanism.

Recent empirical research from Western and English-speaking higher-education settings also suggests that the opportunities and risks of student GenAI use are not unique to China, Thailand-linked contexts. In the UK, survey evidence indicates that many students have already used or considered using generative AI for academic purposes, while levels of acceptance differ across tasks, with more support for assistive uses than for using ChatGPT to produce a whole essay (Johnston et al., 2024). Another UK study likewise identified grammar correction and idea generation as common uses, alongside recurring concerns about plagiarism, privacy, and unclear institutional guidance (Arowosegbe et al., 2024). In Australia, Gruenhagen et al. (2024) found that more than a third of surveyed university students had used a chatbot to assist with assessments and did not necessarily regard such use as an academic-integrity breach. Qualitative evidence from the UK further suggests that students may value ChatGPT for immediacy, reassurance, efficiency, and creativity, while also expressing concerns about equity, peer misuse, and weak policy transparency (Holland & Ciachir, 2025). Taken together, these studies indicate that the benefit-risk tension surrounding GenAI-supported learning is visible beyond the specific regional setting examined in this study. However, they do not yet establish that the same constructs, measurements, or mechanisms operate equivalently across contexts.

1.3. Research questions and hypotheses

Guided by Self-Determination Theory (SDT) and habit-formation frameworks, this study investigates how students' prior motivational climate, particularly high-school autonomy support or pressure, is

associated with their motivational orientation, digital learning behaviors, and AI-use outcomes in university settings. We model two distinct behavioral trajectories in AI learning: The Adaptive Pathway, marked by autonomous motivation and planned trigger readiness, associated with high-quality AI learning and stronger academic functioning. The Compulsive Pathway, marked by controlled motivation and anxiety-based triggers, is associated with overuse, study-life conflict, and diminished well-being.

In addition to the broader global context, this study was also prompted by recurring classroom observations in Thai international higher-education settings. A particularly salient pattern has emerged among Chinese students who completed mainland Chinese high school and then continued into undergraduate programs in Thailand. In internationalized classrooms, these students often show comparatively weaker participation in discussion-based activities and spontaneous academic communication, yet their written assignments frequently display strong and sometimes unedited traces of generative AI use. Typical markers include template-like phrasing, stylistic discontinuities, residual formatting such as double asterisks, excessive dash sequences, or other artifacts that suggest direct pasting of AI outputs with minimal revision. While these observations do not imply that AI use is inherently problematic, they do raise a concrete pedagogical concern: under what motivational and contextual conditions does generative AI function as a constructive scaffold for learning, and when does it drift toward an unreflective shortcut that may hinder students' engagement, confidence, and communicative development? Building on this concern, we address the following research questions (RQs):

RQ1. To what extent are High-School Pressure (HSP), High-School Autonomy Support (HSAS), and Current University Autonomy Support (CAS) associated with Autonomous Motivation (AM), Controlled Motivation (CM), and Planned Trigger Readiness (PTR) for AI-supported learning?

RQ2. Are AM and PTR associated with greater adaptive AI use (ADP), and is CM associated with greater compulsive AI use (COP)?

RQ3. Do ADP and COP differentially relate to behavioral functioning, such that ADP is associated with higher perceived academic performance (APerf) and subjective well-being (WB), whereas COP is associated with lower WB, and are APerf and WB also associated with perceived AI-supported learning quality (AIQ)?

RQ4. Are these structural paths invariant across high-school background groups (China-HS group vs non-China-HS group), and where do key path magnitudes differ?

RQ5 (secondary). After controlling for high-school background, do the CM → COP path and the total indirect effect of CAS → AM → PTR → ADP differ by current university location (China-Uni group vs Thailand-Uni group)?

To address these questions, we tested the hypotheses summarized in Table 1A using structural equation modeling (SEM). The table lists all hypothesized paths, expected directions, and the planned cross-group comparisons.

1.4. Contribution

Theoretically, this study extends Self-Determination Theory by using it to interpret how students' formative motivational experiences may be associated with their current technology-mediated learning behaviors. While prior SDT research has emphasized classroom engagement and academic outcomes, the present model examines how these motivational foundations may be linked to students' interaction with AI, particularly whether their engagement is more consistent with adaptive, goal-directed strategies or with compulsive, trigger-driven overuse. The integration of SDT with the Hook behavior model also offers a novel conceptual lens for examining the intersection of psychological motivation and habit-formation processes in digital education contexts.

Table 1A

Hypotheses and expected directions in the dual-pathway model.

Block	Hypothesis	Path(s)	Expected direction/note
Distal & current predictors → motivation/readiness	H1a	HSP → CM	+
		HSP → PTR	–
	H1b	HSAS → AM	+
		HSAS → PTR	+
	H1c	CAS → AM	+
Motivation & readiness → AI-use pathways		CAS → CM	–
		CAS → PTR (via AM)	Indirect + (CAS → AM → PTR)
	H2a	CM → COP	+
	H2b	AM → PTR	+
		PTR → ADP	+
AI-use pathways → behavioral functioning		AM → ADP (via PTR)	Indirect + (AM → PTR → ADP)
	H3a	ADP → APerf	+
		ADP → WB	+
	H3b	COP → WB	–
	H4	APerf → AIQ	+
Behavioral functioning → final outcome		WB → AIQ	+
	H5a (primary; HS background)	CM → COP	Larger in China-HS than non-China-HS
	H5b (primary; HS background)	CAS → AM → PTR → ADP (total indirect)	Larger in non-China-HS than China-HS
	H5c (secondary; Uni location; controlling HS background)	CM → COP; CAS → AM → PTR → ADP (total indirect)	Differs between China-Uni and Thailand-Uni groups

Note. Abbreviations: HSP = High-School Pressure; HSAS = High-School Autonomy Support; CAS = Current University Autonomy Support; CM = Controlled Motivation; AM = Autonomous Motivation; PTR = Planned Trigger Readiness; COP = Compulsive AI Use; ADP = Adaptive AI Use; APerf = Perceived Academic Performance; WB = Subjective Well-being; AIQ = Perceived AI-supported learning quality.

Methodologically, the study applies dual serial mediation modeling using SEM to examine a theoretically specified chain from distal educational predictors (e.g., high-school climate) to AI-related behavioral and perceived learning outcomes. By conducting multi-group comparisons across students from China and Thailand, the study responds to calls for more culturally contextualized EdTech research and evaluates whether the proposed pattern of associations is similar across the sampled educational groupings.

Practically, the findings offer preliminary implications for educators, curriculum designers, and AI platform developers. The results suggest that autonomy-supportive educational environments may be associated with more self-regulated and effective AI learning habits, indicating the potential value of emphasizing choice, competence, and rationale throughout earlier stages of education, not only at the university level. In parallel, platform designers may consider features that support intentional, low-friction, and goal-aligned use rather than more compulsive forms of engagement. In an era where AI is becoming increasingly embedded in students' academic lives, understanding the motivational conditions associated with healthier forms of AI engagement is both timely and important. Within the China, Thailand-linked contexts examined here, this study offers an integrative and culturally contextualized framework for understanding adaptive and reliance-oriented AI use in education.

2. Method

2.1. Participants and procedure

The sample comprised 508 undergraduate students who participated between March and May 2025. Participants completed an anonymous

online questionnaire about their high school and current university experiences in their own time (outside scheduled class sessions). Recruitment was conducted through announcements posted in university classes and learning management systems, and participation was entirely voluntary.

Students were categorized into two sets of comparison groups: (a) high school country, contrasting those who had attended high school in China (China-HS group, $n = 305$) with those whose high school was in another country (non-China-HS group, $n = 203$), and (b) current university location, contrasting students enrolled at universities in China (China-Uni group, $n = 193$) and those enrolled at a university in Thailand (Thailand-Uni group, $n = 315$). Alongside the focal constructs, participants reported limited non-identifying background information (e.g., major area, study year/level, living on campus, weekly study hours, AI-use frequency and experience, and response language), as well as the country of their high school and current university, which were used to define the multi-group subsamples. To reduce identifiability risk and keep the questionnaire concise, we did not collect sensitive demographics such as age and gender.

All procedures complied with international ethical standards for human subjects research and the 2013 revision of the Declaration of Helsinki (World Medical Association, 2013). The study received an ethics exemption determination from the institutional ethics committee at Raffles International College Bangkok (ID: RICB-ETH-2025-019) and was classified as exempt, minimal-risk anonymous survey research under the institution's regulations for research involving human participants. Participation was voluntary and anonymous. At the start of the survey, respondents read an online information statement and provided informed consent before proceeding. The questionnaire took approximately 15 to 20 min to complete.

2.2. Measures

All constructs were measured with multi-item seven-point Likert scales (1 = strongly disagree, 7 = strongly agree) unless otherwise noted. Higher scores indicated higher levels of the corresponding constructs. Measurement sources and example items are described below, and scale reliabilities for the full sample are reported in Table 1.

2.2.1. High school motivational climate (HSP and HSAS)

High school experiences were assessed using items adapted from the Learning Climate Questionnaire (LCQ) and related self-determination theory instruments (Williams & Deci, 1996). High School Pressure (HSP) captured a grade-focused, exam-centered, and comparison-oriented climate (e.g., "In my high school, grades/ranks/-scholarship competition were frequently emphasized"; "Teachers/-school placed strong emphasis on exam scores"). High School Autonomy

Table 1
Descriptive statistics and reliability (full sample).

Variable	M	SD	α
HSP	4.005	1.942	0.970
HSAS	4.143	1.958	0.976
CAS	4.298	1.934	0.967
CM	4.490	1.872	0.942
AM	4.065	1.897	0.953
PTR	4.099	1.910	0.958
COP	4.698	1.825	0.922
ADP	4.701	1.906	0.956
APerf	4.800	1.900	0.954
WB	4.555	1.893	0.951
AIQ	4.342	1.912	0.959

Note. HSP = high school pressure; HSAS = high school autonomy support; CAS = Current University Autonomy Support; CM = controlled motivation; AM = autonomous motivation; PTR = planned trigger readiness; COP = compulsive AI use; ADP = adaptive AI use; APerf = perceived academic performance; WB = well-being; AIQ = perceived AI-supported learning quality.

Support (HSAS) reflected teachers' encouragement of choice, pacing, and multiple ways of demonstrating understanding (e.g., "Teachers allowed choices in assignment topics or formats"; "I could progress at a self-paced speed"). Each subscale included four items, and internal consistencies were high (all α values above 0.90).

2.2.2. Current University Autonomy support (CAS)

Perceived autonomy support in the current university context was measured with four items adapted from the LCQ to reflect autonomy-supportive instruction at the course level (Williams & Deci, 1996). Items asked students to rate the extent to which their courses and instructors provided meaningful options and flexibility this semester (e.g., "My courses/instructors provide choices in tasks or learning paths this semester"; "I am able to adjust pacing or deadline windows when needed this semester"). The Current University Autonomy Support (CAS) scale showed excellent reliability ($\alpha = .967$).

2.2.3. Motivation (autonomous and controlled)

Drawing on self-determination theory (Deci & Ryan, 2000; Ryan & Deci, 2000), motivation for studying was operationalized as two composite constructs. Autonomous Motivation (AM) combined intrinsic and identified regulation items that reflected learning for interest and personal value (e.g., "I study because I am genuinely interested in the material this semester"; "Studying feels meaningful and valuable to me this semester"). Controlled Motivation (CM) combined introjected and external regulation items that reflected pressure, guilt, or external demands (e.g., "I would feel guilty or anxious if I didn't study hard this semester"; "I often feel compelled by external demands or evaluations this semester"). Items were adapted from validated SDT-based scales (Ryan & Deci, 2000; Williams & Deci, 1996) and contextualized for the current semester. Internal consistency was satisfactory ($\alpha = .953$ for AM; $\alpha = .942$ for CM).

2.2.4. AI learning pathways (PTR, ADP, COP)

Three constructs captured students' typical approaches to AI-assisted learning, informed by research on habit formation and the Hook model of user engagement (Lukyanchikova et al., 2023). Planned Trigger Readiness (PTR) assessed the extent to which students deliberately planned when and how to use AI tools for study (e.g., "At set times/places, I generally start studying/using AI this semester"; "I usually plan steps before I start this semester"). Adaptive AI Use Pathway (ADP) measured constructive, reflective engagement with AI in ways that deepen understanding and support self-regulation (e.g., "This semester, I use AI to plan steps and decide what I should do next"; "This semester, I check or compare AI results before using them in my work"). Compulsive AI Use Pathway (COP) assessed a tendency to rely on AI for quick, minimally checked answers and final wording (e.g., "This semester, I tend to paste questions into AI to get a ready-to-use answer as quickly as possible"; "This semester, when AI gives a plausible answer, I often move on with minimal checking"; "This semester, I rely on AI to produce final wording with only minor edits"). Each construct was measured with four items. Internal consistency was high for all three scales (all α values above 0.90).

2.2.5. Academic performance (APerf)

Perceived academic performance was measured with four self-report items developed for this study, informed by prior self-description work on academic functioning in late adolescence (Marsh & O'Neill, 1984). Items asked students to evaluate how effectively they managed their coursework this semester (e.g., "This semester, I usually complete my coursework on time"; "This semester, I keep up with most academic requirements in my classes"; "This semester, the quality of my academic work generally meets instructors' expectations"; "This semester, I manage my studies effectively overall"). Responses were given on the same seven-point scale, and reliability was high ($\alpha = .954$).

2.2.6. Subjective well-being (WB)

Subjective well-being this semester was assessed with four items developed for this study, focusing on students' feelings of energy, study-life balance, sleep/rest, and workable stress management (e.g., "This semester, I generally feel positive and energetic"; "This semester, I maintain a workable balance between study and life"; "This semester, I can manage study-related stress in a workable way"). Items were rated on the same seven-point scale, and scores were averaged, with higher scores indicating higher subjective well-being. The scale showed excellent internal consistency ($\alpha = .951$).

2.2.7. Perceived AI-supported learning quality (AIQ)

Perceived AI-supported learning quality (AIQ) was operationalized as a self-reported indicator of students' evaluations of the quality of AI-supported learning in the current semester, rather than as an objective measure of learning gains. In this study, AIQ refers to students' subjective evaluations of the perceived quality and usefulness of AI-supported learning in the current semester, rather than to objective learning gains or externally verified academic performance.

The four items were developed for this study to capture whether students perceived that GenAI use improved the depth, reliability, and efficiency of their learning (e.g., "This semester, using AI helps me deepen my understanding"; "This semester, AI outputs are useable and can be organized into reliable study materials"; "This semester, after checking, I am generally confident in AI outputs' accuracy/consistency"; "This semester, AI helps me work more efficiently and with more assurance"). Given the cross-sectional, anonymous, multi-institution survey design, a perceived outcome measure was used as a pragmatic indicator of students' learning-related evaluations of GenAI use, while acknowledging that it does not substitute for direct performance-based assessment. Responses were made on the same seven-point scale. Internal consistency for the AIQ scale was excellent ($\alpha = .959$).

2.2.8. Sampling and recruitment

Data were collected via a cross-sectional online survey administered to undergraduate students in China and Thailand. We used a convenience and volunteer sampling strategy, with targeted recruitment in both countries to capture variation in high-school background and current university context. Participation was voluntary and uncompensated, and all responses were completed outside of scheduled class time.

On the Chinese side, the survey link was distributed through two main channels: (a) a recruitment post on the social-media platform Rednote and (b) internal communication channels at five universities in Jiangsu Province that preferred not to be named. These institutions included both public and private universities (4 public, 1 private). In all cases, instructors shared the survey link with their students, but data collection took place outside class sessions and was not tied to course grades or attendance.

On the Thailand side, participants were recruited from four universities. Two private institutions consented to be named in this article, Siam University and Raffles International College Bangkok, whereas two public universities requested anonymity. As in China, instructors circulated the survey link to their students, who could decide whether to participate in their own time. This procedure yielded a diverse university sample spanning business and related majors in both Chinese and Thai higher education.

To enhance data quality, the final block of the questionnaire (Block 9: Quality Control and Debrief) contained two attention-check items. In the first item, participants were instructed to select the option "Agree" on a standard 7-point Likert scale to confirm that they were reading the questions carefully. In the second item, they were asked to select the number "5" from a 1-to-7 numeric response scale. Participants who failed at least one of these attention checks were excluded from the analytical sample (33 respondents), resulting in a final sample of 508 students for the SEM analyses.

This study used a cross-sectional survey with retrospective reports of high-school experiences; therefore, the SEM paths should be interpreted as theoretically motivated associations rather than evidence of temporal or causal effects.

2.2.9. Data analysis plan

All analyses were conducted in SPSS 29 and AMOS 29. Data screening confirmed that item-level skewness and kurtosis values were within acceptable ranges (absolute skewness less than 2 and absolute kurtosis less than 7). Missing data were below 5 percent on all items, and Little's MCAR test indicated that missingness was random (p greater than 0.05). Given the sample size ($N = 508$) and model complexity, the data set was adequate for structural equation modeling (Kline, 2016; Little, 1988).

Following the two-step approach to structural equation modeling (Anderson & Gerbing, 1988), analyses proceeded in several stages. First, descriptive statistics, Cronbach alpha values, and Pearson correlations were inspected to evaluate data quality and initial relationships among constructs.

Second, a series of confirmatory factor analyses (CFA) tested the hypothesized measurement model for all latent constructs. Items were specified to load on their intended factors only, and residuals were not allowed to correlate unless there was a strong theoretical justification. Standardized factor loadings were expected to exceed 0.50. Convergent validity was evaluated using composite reliability (CR, with a target of 0.70 or higher) and average variance extracted (AVE, with a target of 0.50 or higher). Discriminant validity was evaluated using the Fornell and Larcker (1981) criterion that each construct's AVE should exceed the squared correlations with other constructs.

Third, overall model fit was evaluated using multiple indices: the chi-square statistic divided by degrees of freedom (chi-square/df, with values below 3.0 indicating acceptable fit), the Comparative Fit Index (CFI), the Tucker-Lewis Index (TLI), the Root Mean Square Error of Approximation (RMSEA) with 90 percent confidence interval, and the Standardized Root Mean Square Residual (SRMR). Model fit was considered acceptable when CFI and TLI were at least 0.90 (good if 0.95 or higher), RMSEA was at or below 0.08 (good at or below 0.06), and SRMR was at or below 0.08 (Hu & Bentler, 1999). Although the chi-square statistic was reported for completeness, fit evaluation emphasized the incremental and residual indices, which are less sensitive to sample size.

Fourth, after a satisfactory measurement fit was established, the hypothesized structural model was estimated to test the proposed dual-pathway relations among high-school and university motivational climate, motivational quality, AI-use pathways, academic performance, well-being, and perceived AI-supported learning quality. Direct, indirect, and total effects were examined, and mediation was tested using bias-corrected bootstrapping with 5,000 resamples. Indirect effects were considered statistically significant when the 95 percent confidence interval did not include zero (Preacher & Hayes, 2008). Fig. 1 presents the hypothesized dual-pathway structural model that was evaluated in these analyses.

Fifth, multi-group structural equation modeling examined whether measurement and structural relations were equivalent across the high-school and university subgroups (China-HS group versus non-China-HS group, and China-Uni group versus Thailand-Uni group). Measurement invariance was assessed in a hierarchical sequence of configural, metric, and scalar models (Wu et al., 2007). Nested model comparisons used Satorra-Bentler scaled chi-square difference tests and changes in CFI and RMSEA (Satorra & Bentler, 2001). Following recommendations by Cheung and Rensvold (2002) and Chen (2007), invariance was judged to hold when the change in CFI was 0.010 or less, and the change in RMSEA was 0.015 or less. When full scalar invariance could not be achieved, partial invariance models were specified by freeing non-invariant parameters. After establishing an acceptable level of invariance, latent means and key structural path coefficients were compared

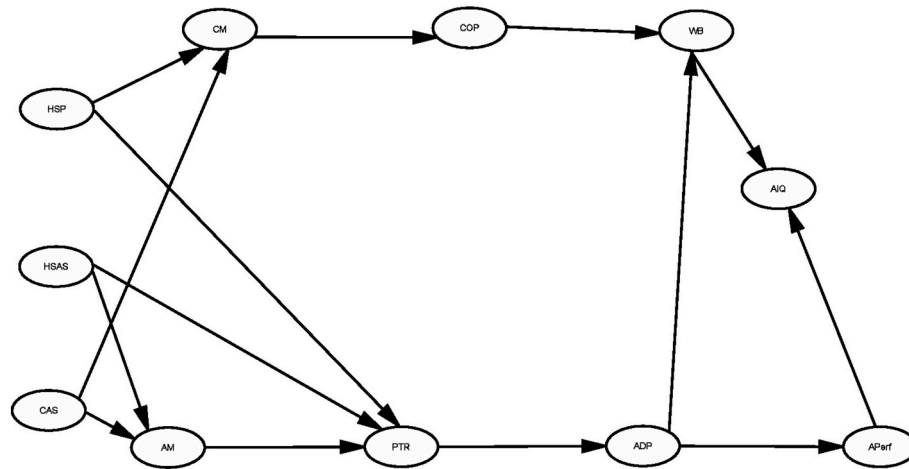


Fig. 1. Hypothesized dual-pathway model of high-school and university motivational climate, AI-learning pathways, and student outcomes.

Note. HSP = High-School Pressure; HSAS = High-School Autonomy Support; CAS = Current University Autonomy Support; AM = Autonomous Motivation; CM = Controlled Motivation; PTR = Planned Trigger Readiness; ADP = Adaptive AI Use; COP = Compulsive AI Use; APerf = Perceived Academic Performance; WB = Subjective Well-being; AIQ = Perceived AI-supported learning quality.

across groups.

This study was reported in accordance with the STROBE guidelines for cross-sectional studies, and the completed checklist is included with the submission.

3. Results

3.1. Descriptive statistics and measurement model

Table 1 reports the means, standard deviations, and reliabilities for all study variables in the full sample. On the seven-point scales, mean scores ranged from about 4.05 to 4.800, indicating generally moderate to high perceived high-school pressure (HSP), high-school autonomy support (HSAS), Current University Autonomy Support (CAS), dual motivational qualities (CM, AM), AI-use pathways (PTR, COP, ADP), and outcomes (APerf, WB, AIQ). Standard deviations were close to 2.0 for all constructs, suggesting adequate variability. Cronbach's alpha coefficients ranged from 0.922 to 0.976, demonstrating excellent internal consistency across all scales.

Zero-order correlations for the full sample are shown in Table 2. The correlational pattern was consistent with the proposed dual-pathway logic: high-school pressure was positively associated with controlled motivation and compulsive AI use, and negatively associated with planned trigger readiness, adaptive AI use, and downstream outcomes. In contrast, high-school autonomy support and current university autonomy support were positively associated with autonomous motivation, planned trigger readiness, adaptive AI use, and perceived outcomes. Detailed coefficients are reported in Table 2.

The measurement properties of the eleven latent constructs were examined using CFA. Standardized factor loadings were consistently high (all $\lambda > 0.89$), and the constructs demonstrated strong reliability and convergent validity (CR and AVE). Discriminant validity was supported based on AVE-based criteria. Detailed results are provided in Appendix Table A1.

3.2. Structural model, full sample

The hypothesized structural model was estimated for the full sample. Model fit indices are reported in Table 3, and standardized structural coefficients are presented in Fig. 2. Overall, the results were consistent with the proposed dual-pathway framework: a pressure-compulsive route linked higher perceived high-school pressure to higher

Table 3

Fit indices for the full-sample structural model (N = 508).

Fit index	Value
CMIN/df	1.61
CFI	0.981
TLI	0.979
RMSEA	0.035

Note. CMIN/df = chi-square divided by degrees of freedom; CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation.

Table 2

Correlations among study variables (full sample).

	HSP	HSAS	CAS	CM	AM	PTR	COP	ADP	APerf	WB	AIQ
HSP	1.000	0.049	0.015	0.531	0.044	-0.403	0.341	-0.168	-0.199	-0.133	-0.152
HSAS	0.049	1.000	-0.020	0.070	0.626	0.527	0.005	0.551	0.280	0.306	0.221
CAS	0.015	-0.020	1.000	-0.594	0.512	0.409	-0.463	0.462	0.471	0.403	0.432
CM	0.531	0.070	-0.594	1.000	-0.265	-0.482	0.674	-0.361	-0.454	-0.422	-0.430
AM	0.044	0.626	0.512	-0.265	1.000	0.611	-0.251	0.773	0.542	0.514	0.443
PTR	-0.403	0.527	0.409	-0.482	0.611	1.000	-0.371	0.718	0.559	0.457	0.440
COP	0.341	0.005	-0.463	0.674	-0.251	-0.371	1.000	-0.263	-0.488	-0.425	-0.420
ADP	-0.168	0.551	0.462	-0.361	0.773	0.718	-0.263	1.000	0.633	0.560	0.507
APerf	-0.199	0.280	0.471	-0.454	0.542	0.559	-0.488	0.633	1.000	0.461	0.620
WB	-0.133	0.306	0.403	-0.422	0.514	0.457	-0.425	0.560	0.461	1.000	0.607
AIQ	-0.152	0.221	0.432	-0.430	0.443	0.440	-0.420	0.507	0.620	0.607	1.000

Note. Values are Pearson correlations. See Table 1 for variable labels.

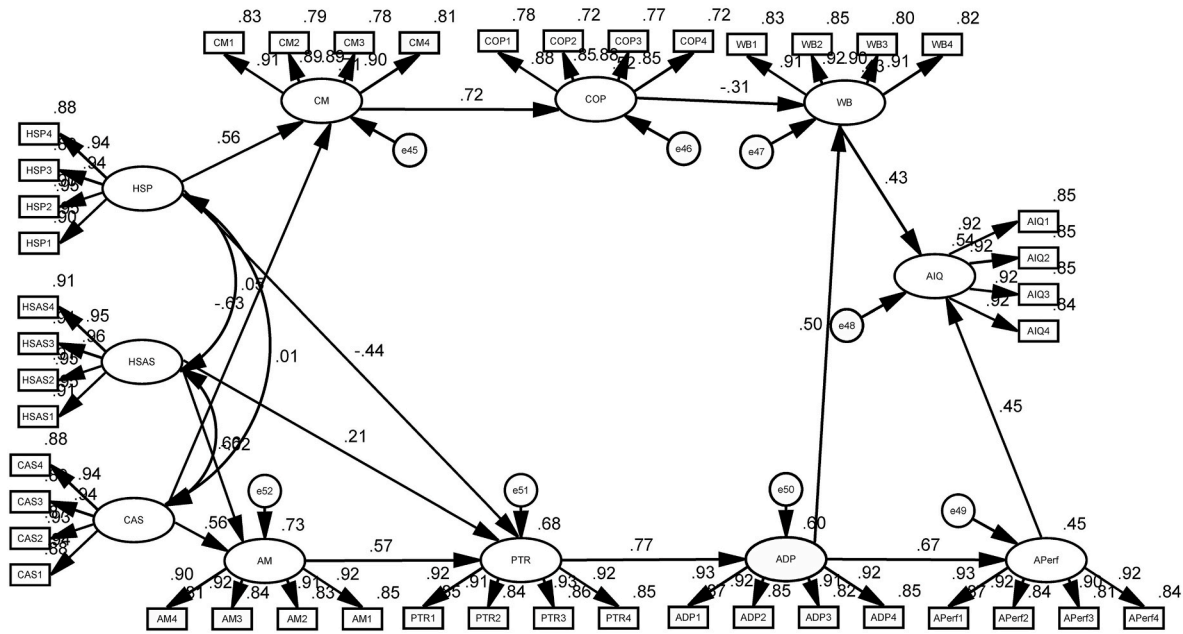


Fig. 2. Structural equation model of dual AI learning pathways for the full sample (N = 508).

Note. Standardized path coefficients are shown. For clarity, measurement indicators are omitted (see Appendix Table A1). HSP = High-School Pressure; HSAS = High-School Autonomy Support; CAS = Current University Autonomy Support; CM = Controlled Motivation; AM = Autonomous Motivation; PTR = Planned Trigger Readiness; COP = Compulsive AI Use; ADP = Adaptive AI Use; APerf = Perceived Academic Performance; WB = Subjective Well-being; AIQ = Perceived AI-supported learning quality.

controlled motivation, more compulsive AI use, and lower well-being, whereas an autonomy-adaptive route linked higher perceived high-school autonomy support and Current University Autonomy Support to higher autonomous motivation, greater planned trigger readiness, and more adaptive AI use. In turn, adaptive AI use was positively associated with perceived academic performance, well-being, and perceived AI-supported learning quality, whereas compulsive AI use was negatively associated with well-being.

3.2.1. Mediation results, full sample

Mediation was tested using bias-corrected bootstrapping with 5,000 resamples. As summarized in Table 4, all focal indirect effects were statistically significant. Higher perceived high-school pressure showed a negative indirect association with perceived AI-supported learning quality through the pressure-compulsive route, whereas high-school autonomy support showed positive indirect associations with perceived AI-supported learning quality through the autonomy-adaptive route, via planned trigger readiness, adaptive AI use, and links through perceived academic performance and well-being. Together, these indirect effects were consistent with the proposed dual-pathway interpretation.

Table 4

Bootstrapped indirect effects in the full structural model (N = 508).

Predictor	Mediator chain	Outcome	Estimate	SE	95% CI lower	95% CI upper	p
HSP	CM → COP → WB	AIQ	−0.088	0.013	−0.115	−0.066	<0.001
HSAS	PTR → ADP → APerf	AIQ	0.052	0.012	0.031	0.077	<0.001
HSAS	AM → PTR → ADP → APerf	AIQ	0.094	0.013	0.071	0.124	<0.001
HSAS	AM → PTR → ADP → WB	AIQ	0.063	0.010	0.047	0.085	<0.001

Note. Estimates are unstandardized indirect effects with bias-corrected 95 percent confidence intervals based on 5,000 bootstrap samples. HSP = High-School Pressure; HSAS = High-School Autonomy Support; CM = Controlled Motivation; COP = Compulsive AI Use; AM = Autonomous Motivation; PTR = Planned Trigger Readiness; ADP = Adaptive AI Use; APerf = Perceived Academic Performance; WB = Subjective Well-being; AIQ = Perceived AI-Supported Learning Quality.

3.3. Measurement invariance across high school and university groups

Before comparing structural relations across groups, we tested measurement invariance across (a) high-school background (China-HS group vs. non-China-HS group) and (b) current university location (China-Uni group vs. Thailand-Uni group) using multi-group CFAs. Using commonly adopted thresholds ($\Delta CFI \leq 0.010$; $\Delta RMSEA \leq 0.015$), results supported scalar-level invariance for both grouping schemes (all $\Delta CFI \leq 0.004$; all $\Delta RMSEA \leq 0.001$). Detailed fit statistics for configural, metric, and scalar models are reported in Appendix Table A2, supporting meaningful cross-group comparisons of structural paths.

Table 7

Model fit indices for structural models by high school background (China-HS group vs. non-China-HS group).

Group	CMIN/df	CFI	TLI	RMSEA
China-HS group (N = 305)	1.39	0.980	0.978	0.036
non-China-HS group (N = 203)	1.21	0.983	0.982	0.032

Note. CMIN/df = chi-square divided by degrees of freedom; CFI = Comparative Fit Index; TLI = Tucker–Lewis Index; RMSEA = Root Mean Square Error of Approximation. All values indicate good model fit for both groups.

3.3.1. High school groups (China-HS group vs. non-China-HS group)

For the high-school background comparison, the structural model fit the data well in both groups (Table 7; Figs. 3–4). Across the China-HS and non-China-HS groups, the overall pattern of relations closely resembled the full-sample model (Table 8): autonomy-supportive climates were associated with a more autonomous and planful adaptive-AI route, whereas high-school pressure was associated with controlled motivation and a more compulsive route. Between-group parameter difference tests indicated that only one structural path differed significantly across groups: the CM → COP association was stronger in the China-HS group than in the non-China-HS group. All other path differences were not statistically significant.

Across both groups, the models explained substantial variance in the key endogenous variables, indicating that the dual-pathway framework accounted for meaningful proportions of variance in both adaptive and compulsive AI use, as well as in downstream perceived outcomes, across high-school backgrounds.

To address H5, we estimated bootstrapped mediation models separately for the China-HS and non-China-HS groups (Table 9). In both groups, high-school pressure showed significant negative indirect associations with perceived AI-supported learning quality through both the pressure-compulsive route and the reduced-adaptive-engagement route. In contrast, high-school autonomy support showed significant positive indirect associations with perceived AI-supported learning quality through the autonomy-adaptive route, including both performance-based and well-being-based chains. Although differences between group-specific indirect effects were not formally tested, HSAS-related indirect estimates were numerically larger in the non-China-HS group.

3.4. University groups (China-uni group vs. Thailand-uni group)

The structural model showed good fit in both university subsamples (Table 10; Figs. 5–6), indicating that the proposed dual-pathway structure fit both sampled university groupings adequately.

Across the two university groups, the overall pattern of associations closely resembled the full-sample model (Table 11). In both groups,

autonomy-supportive climates were associated with a more autonomous and planful adaptive-AI route, whereas pressure-related experiences were associated with a more controlled and compulsive route linked to lower well-being and indirectly to lower perceived AI-supported learning quality.

Between-group parameter difference tests indicated that only one structural path differed significantly across university subsamples (Table 11): the CM → COP association was stronger in the China-uni group than in the Thailand-uni group. All other path differences were not statistically significant, and explained variance for the key endogenous variables was substantial in both groups (Figs. 5–6).

3.5. Additional multi-group mediation for university groups

To further address RQ5 and H5c, we estimated bias-corrected bootstrapped indirect effects separately for the China-uni and Thailand-uni groups (Table 12). Across both university contexts, the indirect-effect patterns were consistent with the proposed dual-pathway framework. Specifically, higher perceived high-school pressure showed negative indirect associations with perceived AI-supported learning quality through the pressured-compulsive chain (via controlled motivation, compulsive AI use, and lower well-being), and also through the readiness-adaptation route (via reduced planned trigger readiness, lower adaptive AI use, and downstream outcomes).

In parallel, high-school autonomy support exhibited positive indirect associations with perceived AI-supported learning quality through the autonomy-adaptive pathway. Indirect effects involving planned trigger readiness and adaptive AI use were consistently supported, including routes operating through perceived academic performance and well-being. Taken together, these bootstrapped mediation results provide convergent evidence that the theorized pressure-linked and autonomy-linked mechanisms generalize across students in Chinese and Thai university settings (Table 12).

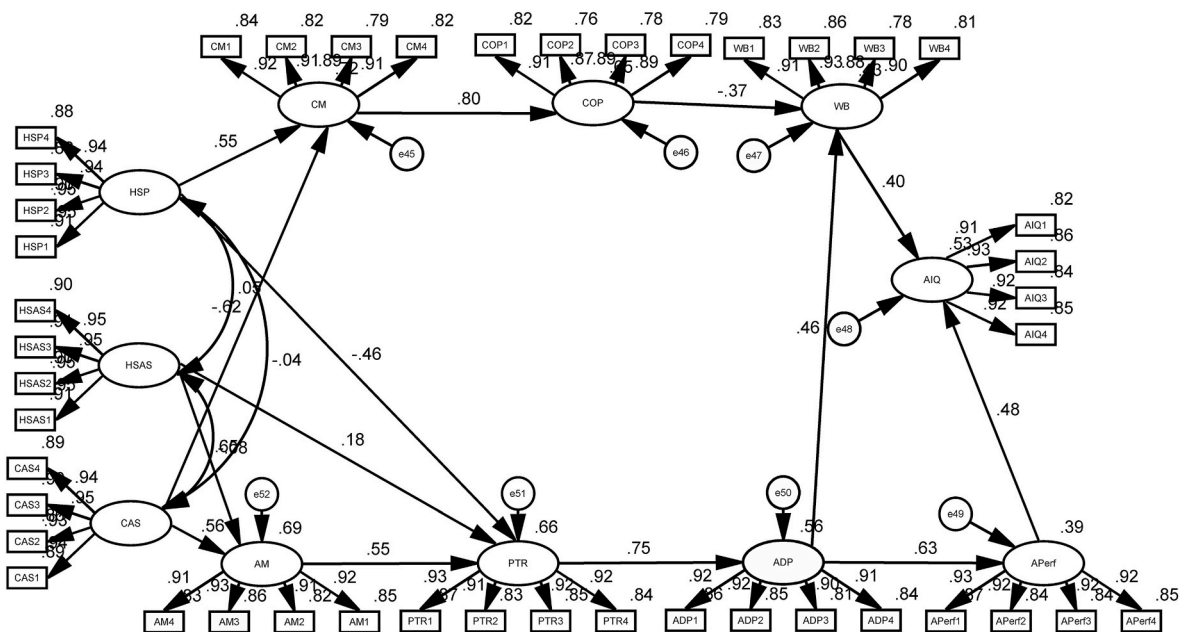


Fig. 3. Structural model for students who attended high school in China (China-HS group).

Note. HSP = high school pressure; HSAS = high school autonomy support; CAS = Current University Autonomy Support; CM = controlled motivation; AM = autonomous motivation; PTR = planned trigger readiness; COP = compulsive AI use; ADP = adaptive AI use; APerf = perceived academic performance; WB = well-being; AIQ = perceived AI-supported learning quality. Values on the arrows are standardized path coefficients (β). Values in the circles/rectangles next to endogenous variables represent R². All paths shown are statistically significant at $p < .001$. Model fit: CMIN/df = 1.39, CFI = 0.980, TLI = 0.978, RMSEA = 0.036.

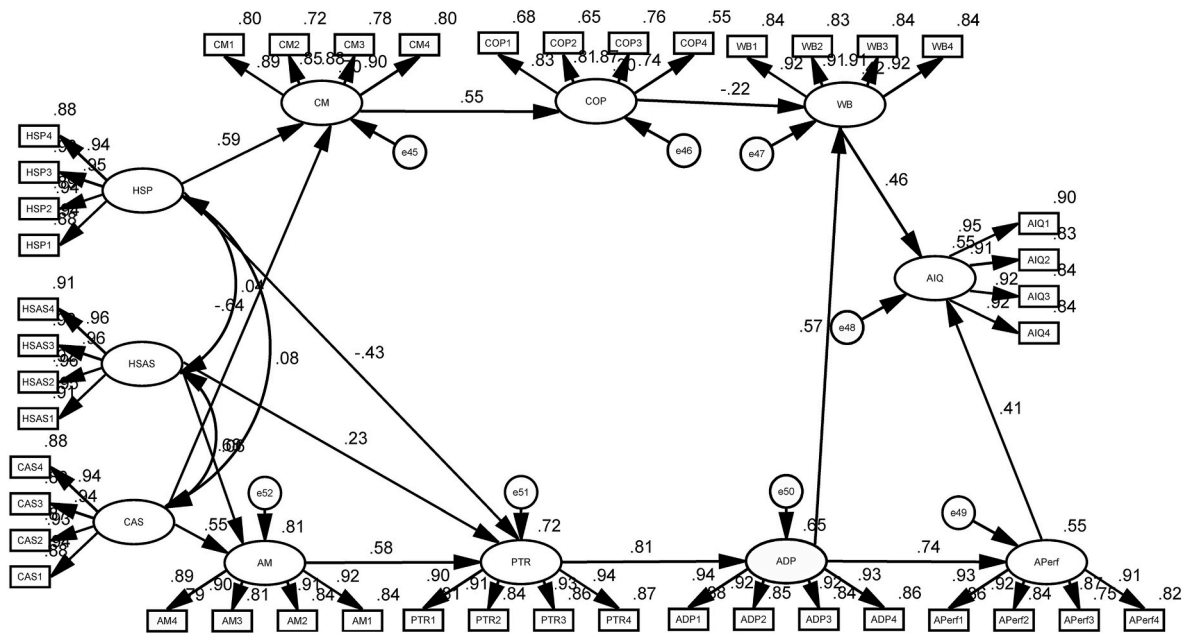


Fig. 4. Structural model for students who attended non-Chinese high schools (non-China-HS group)

Note. HSP = high school pressure; HSAS = high school autonomy support; CAS = Current University Autonomy Support; CM = controlled motivation; AM = autonomous motivation; PTR = planned trigger readiness; COP = compulsive AI use; ADP = adaptive AI use; APerf = perceived academic performance; WB = well-being; AIQ = perceived AI-supported learning quality. Values on the arrows are standardized path coefficients (β). Values in the circles/rectangles next to endogenous variables represent R^2 . All paths shown are statistically significant at $p < .001$. Model fit: CMIN/df = 1.21, CFI = 0.983, TLI = 0.982, RMSEA = 0.032.

Table 8

Structural path coefficients for high-school background groups and between-group difference tests (China-HS group vs. non-China-HS group).

Path	China-HS β	non-China-HS β	$\Delta\beta$ (China–non)	z(diff)	p(diff)	Sig diff.
HSAS $\rightarrow$ AM	0.650	0.680	–0.030	–0.194	0.846	No
CAS $\rightarrow$ AM	0.561	0.564	–0.003	0.394	0.694	No
HSP $\rightarrow$ CM	0.551	0.591	–0.040	–0.100	0.920	No
CAS $\rightarrow$ CM	–0.621	–0.644	+0.023	–0.688	0.492	No
AM $\rightarrow$ PTR	0.553	0.584	–0.031	–0.775	0.439	No
HSP $\rightarrow$ PTR	–0.463	–0.434	–0.029	0.305	0.760	No
HSAS $\rightarrow$ PTR	0.181	0.230	–0.049	–0.725	0.468	No
PTR $\rightarrow$ ADP	0.750	0.810	–0.060	–0.665	0.506	No
ADP $\rightarrow$ APerf	0.633	0.744	–0.111	–0.221	0.825	No
ADP $\rightarrow$ WB	0.464	0.570	–0.106	–1.442	0.149	No
CM $\rightarrow$ COP	.801	.554	+.247	3.982	< .001	Yes***($p < .001$)
COP $\rightarrow$ WB	–0.370	–0.222	–0.148	–0.779	0.436	No
APerf $\rightarrow$ AIQ	0.481	0.414	+0.067	–0.072	0.943	No
WB $\rightarrow$ AIQ	0.401	0.464	–0.063	–0.883	0.377	No

Note. β values are standardized path coefficients. $\Delta\beta = \beta(\text{China-HS}) - \beta(\text{non-China-HS})$. $z(\text{diff})$ and $p(\text{diff})$ are based on between-group parameter difference tests computed from unstandardized estimates and their standard errors. Rows with $p(\text{diff}) < 0.05$ are highlighted (bolded or shaded) to indicate significant between-group differences.

4. Discussion

The present study aimed to clarify the psychological processes potentially associated with university students' interactions with generative artificial intelligence (GenAI) for learning. By integrating Self-Determination Theory (SDT) with the Hook model of habit formation (Eyal, 2014), the study tested a dual-pathway model linking students' retrospective reports of high-school learning climate (pressure vs. autonomy support) to current motivational orientations, AI-use patterns, and perceived outcomes. Because the study used a cross-sectional survey design with retrospective reports of high-school experiences, the directional paths specified in the SEM should be interpreted as theoretically motivated associations rather than as evidence of temporal or causal effects (Deci & Ryan, 2000; Hu et al., 2025; Lukyanchikova et al., 2023; Ryan & Deci, 2000, 2020).

Overall, the results were consistent with two distinguishable patterns

of association. First, higher perceived high-school pressure was associated with higher controlled motivation, which in turn was associated with more compulsive AI use and lower well-being; these associations were further linked to lower perceived AI-supported learning quality. Second, higher perceived high-school autonomy support and current university autonomy support were associated with higher autonomous motivation and greater planned trigger readiness, which were linked to more adaptive AI use and, in turn, to higher perceived academic performance, higher well-being, and higher perceived AI-supported learning quality. Consistent with this interpretation, the reported bootstrapped indirect effects provide additional support for the indirect paths specified in the model (see Results Section 3.2.1 and Table 4 for the full sample, and Tables 9 and 12 for subgroup analyses).

Before interpreting these modeled patterns further, it is important to note that the high-school climate variables were measured through retrospective self-reports and may therefore be shaped partly by

Table 9

Bootstrapped indirect effects for key pathways by high-school background (China-HS group vs. non-China-HS group).

Indirect path	China-HS group Estimate	China-HS group SE	China-HS group 95% CI lower	China-HS group 95% CI upper	non-China-HS group Estimate	non-China-HS group SE	non-China-HS group 95% CI lower	non-China-HS group 95% CI upper
HSP → CM → COP → WB → AIQ	-0.066	0.015	-0.099	-0.041	-0.033	0.012	-0.063	-0.014
HSP → PTR → ADP → APerf → AIQ	-0.104	0.015	-0.138	-0.077	-0.106	0.023	-0.157	-0.067
HSAS → PTR → ADP → APerf → AIQ	0.041	0.013	0.019	0.070	0.056	0.019	0.024	0.102
HSAS → AM → PTR → ADP → APerf → AIQ	0.082	0.015	0.057	0.116	0.097	0.024	0.058	0.152
HSP → PTR → ADP → WB → AIQ	-0.063	0.012	-0.089	-0.043	-0.091	0.018	-0.131	-0.059
HSAS → PTR → ADP → WB → AIQ	0.025	0.008	0.012	0.043	0.048	0.018	0.019	0.091
HSAS → AM → PTR → ADP → WB → AIQ	0.050	0.010	0.033	0.074	0.084	0.019	0.053	0.129

Note. Estimates are bias-corrected bootstrapped indirect effects based on 5,000 resamples. All confidence intervals exclude zero, indicating statistically significant indirect effects (all $p < .01$). HSP = high-school pressure; HSAS = high-school autonomy support; AM = autonomous motivation; CM = controlled motivation; PTR = planned trigger readiness; ADP = adaptive AI use; COP = compulsive AI use; APerf = perceived academic performance; WB = well-being; AIQ = perceived AI-supported learning quality.

Table 10

Structural model fit indices for university subsamples.

Group	CMIN/df	CFI	TLI	RMSEA
China-Uni group (n = 193)	1.295	0.976	0.974	0.039
Thailand-Uni group (n = 315)	1.318	0.984	0.983	0.032

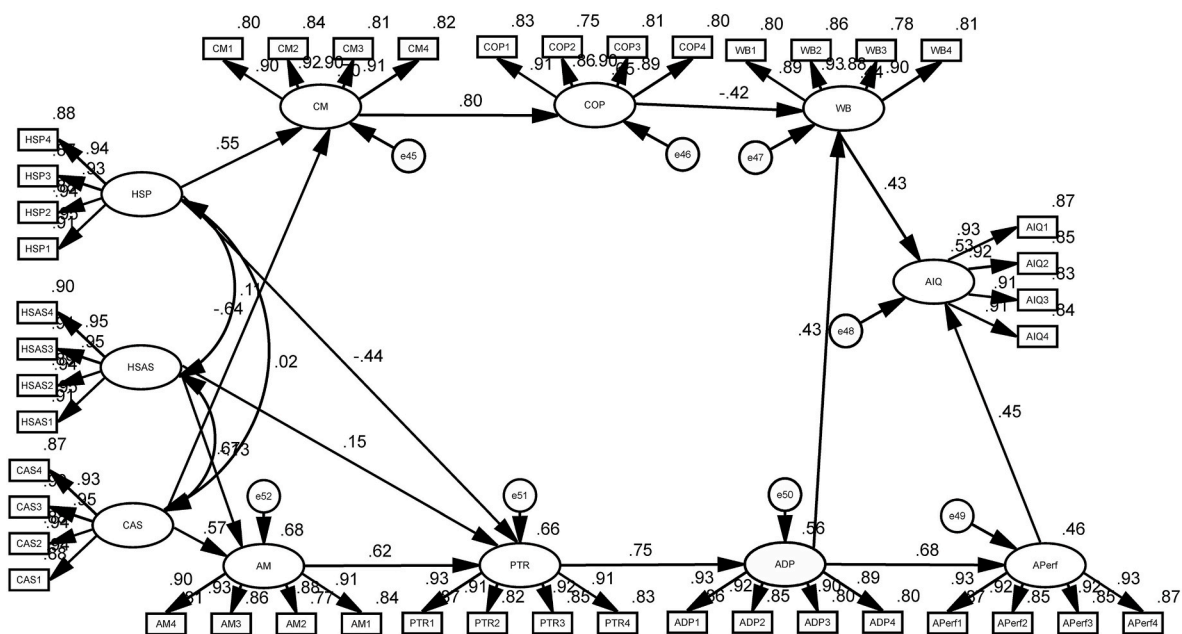
students' current psychological and academic states. For example, students with lower current well-being, higher controlled motivation, or more reliance-oriented AI-use patterns may be more likely to recall their prior high-school environment as pressure-laden, whereas students with more adaptive functioning may recall greater autonomy support. Accordingly, the observed structural paths should be interpreted as

theory-consistent associations rather than as evidence that past high-school climate temporally produced current AI-learning patterns.

Throughout the Discussion, terms such as “pathway” and “route” are used as shorthand for theoretically specified patterns of association within the SEM, rather than as evidence of confirmed temporal sequences.

4.1. Psychological correlates of reliance-oriented AI use

A primary contribution of this study to the behavioral psychology literature is the empirical distinction between adaptive engagement and compulsive dependence within a single, theoretically specified framework. Our findings regarding the compulsive-use pattern align with recent conceptualizations of “GenAI dependency,” which distinguishes

**Fig. 5.** Structural model for students enrolled at Chinese universities (China-Uni group).

Note. HSP = high-school pressure; HSAS = high-school autonomy support; CAS = Current University Autonomy Support; CM = controlled motivation; AM = autonomous motivation; PTR = planned trigger readiness; COP = compulsive AI use; ADP = adaptive AI use; APerf = perceived academic performance; WB = well-being; AIQ = perceived AI-supported learning quality. Values on the arrows are standardized path coefficients (β). Values next to endogenous variables represent R^2 . All paths shown are statistically significant at $p < .001$. Model fit: CMIN/df = 1.295, CFI = 0.976, TLI = 0.974, RMSEA = 0.039.

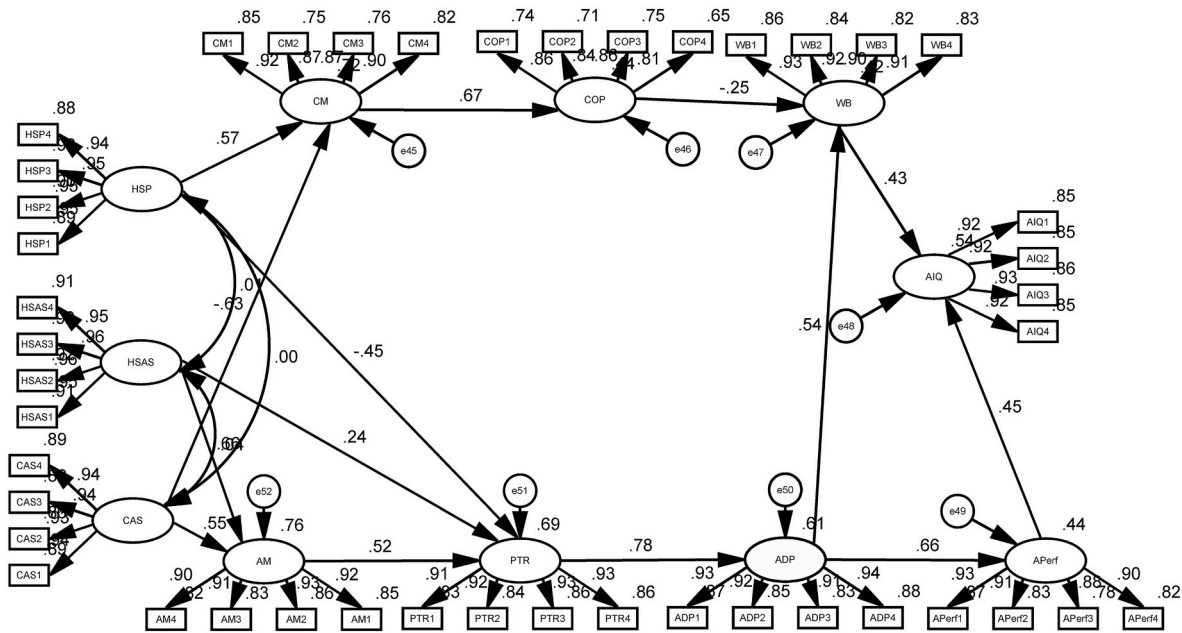


Fig. 6. Structural model for students enrolled at Thai universities (Thailand-Uni group)

Note. HSP = high-school pressure; HSAS = high-school autonomy support; CAS = Current University Autonomy Support; CM = controlled motivation; AM = autonomous motivation; PTR = planned trigger readiness; COP = compulsive AI use; ADP = adaptive AI use; APerf = perceived academic performance; WB = well-being; AIQ = perceived AI-supported learning quality. Values on the arrows are standardized path coefficients (β). Values next to endogenous variables represent R^2 . All paths shown are statistically significant at $p < .001$. Model fit: CMIN/df = 1.318, CFI = 0.984, TLI = 0.983, RMSEA = 0.032.

Table 11

Structural path coefficients for university subsamples and between-group difference tests.

Path	China-Uni β	Thailand-Uni β	$\Delta\beta$ (China-Uni – Thailand-Uni)	$z(\text{diff})$	$p(\text{diff})$	Sig diff.
HSAS → AM	0.671	0.661	0.010	−0.116	0.907	No
CAS → AM	0.544	0.551	−0.007	−0.116	0.907	No
HSP → CM	0.551	0.573	−0.022	−0.200	0.841	No
CAS → CM	−0.640	−0.631	−0.009	−0.302	0.763	No
AM → PTR	0.620	0.523	0.097	1.033	0.302	No
HSAS → PTR	0.150	0.243	−0.093	−1.424	0.155	No
HSP → PTR	−0.441	−0.450	0.009	0.695	0.487	No
CM → COP	.801	.671	.130	3.296	.001	Yes ($p < .05$)
PTR → ADP	0.750	0.780	−0.030	−0.421	0.674	No
ADP → APerf	0.680	0.661	0.019	1.814	0.070	No
ADP → WB	0.430	0.541	−0.111	−1.500	0.134	No
COP → WB	−0.421	−0.251	−0.170	−0.768	0.443	No
APerf → AIQ	0.450	0.452	−0.002	−0.712	0.476	No
WB → AIQ	0.433	0.430	0.003	0.381	0.703	No

Note. $\Delta\beta$ is based on standardized coefficients. $z(\text{diff})$ and $p(\text{diff})$ are based on between-group parameter difference tests computed from unstandardized estimates and standard errors. Rows with $p < .05$ are highlighted to indicate significant between-group differences.

functional reliance from maladaptive dependence characterized by impaired control (Revesai, 2025). In our model, controlled motivation showed a strong association with compulsive AI use, suggesting that more controlled regulation may co-occur with reliance patterns that are less reflective and more coping-oriented.

This pattern is consistent with the “cognitive offloading” hypothesis, whereby learners under evaluative threat or fear of failure may rely on external tools to reduce immediate cognitive and emotional burden (Lee et al., 2025). Interpreted through a habit-formation lens, the “trigger” for these learners may often be negative affect (e.g., anxiety or guilt), which increases the likelihood of rapid, minimally checked AI use as a coping response. While the present data cannot establish a causal feedback loop, the indirect effects reported in the results are consistent with the concern that in pressure-oriented climates, students may be more likely to treat AI as an escape valve rather than a cognitive scaffold, and this pattern is associated with lower well-being and lower perceived AI-supported learning quality (Table 4; Tables 9 and 12).

4.2. Autonomy as a protective correlate in digital behavior

Conversely, the adaptive-use pattern highlights the role of motivational quality in more constructive GenAI engagement. The association between autonomous motivation and planned trigger readiness suggests that self-determined learners are more likely to integrate GenAI into learning in a planful and intentional manner, rather than using it reactively under external pressure. This finding extends recent work applying SDT to GenAI, which argues that supporting autonomy is important for self-regulated learning in automated environments (Chiu, 2024).

From the perspective of the Hook model, planned trigger readiness can be interpreted as learners intentionally structuring when and how GenAI is used, rather than being “pulled” into use by situational stress. In educational terms, this distinction implies that the same tool can function differently depending on motivational context: for autonomy-aligned learners, GenAI is more likely to be integrated into an existing

Table 12

Bootstrapped indirect effects for key pathways by university location (China-Uni group vs. Thailand-Uni group).

Indirect path	China-Uni group Estimate	China-Uni group SE	China-Uni group 95% CI lower	China-Uni group 95% CI upper	Thailand-Uni group Estimate	Thailand-Uni group SE	Thailand-Uni group 95% CI lower	Thailand-Uni group 95% CI upper
HSP → CM → COP → WB → AIQ	−0.080	0.019	−0.127	−0.050	−0.041	0.011	−0.067	−0.023
HSP → PTR → ADP → APerf → AIQ	−0.103	0.021	−0.149	−0.067	−0.104	0.016	−0.137	−0.076
HSAS → PTR → ADP → APerf → AIQ	0.035	0.017	0.005	0.073	0.057	0.014	0.032	0.089
HSAS → AM → PTR → ADP → APerf → AIQ	0.097	0.022	0.060	0.149	0.079	0.015	0.053	0.113
HSP → PTR → ADP → WB → AIQ	−0.061	0.016	−0.100	−0.036	−0.082	0.013	−0.110	−0.060
HSAS → PTR → ADP → WB → AIQ	0.021	0.011	0.003	0.046	0.045	0.012	0.025	0.072
HSAS → AM → PTR → ADP → WB → AIQ	0.058	0.016	0.032	0.096	0.062	0.011	0.043	0.090

Note. Estimates are bias-corrected bootstrapped indirect effects based on 5,000 resamples. All confidence intervals exclude zero, indicating statistically significant indirect effects (all $p < .01$). HSP = high-school pressure; HSAS = high-school autonomy support; AM = autonomous motivation; CM = controlled motivation; PTR = planned trigger readiness; ADP = adaptive AI use; COP = compulsive AI use; APerf = perceived academic performance; WB = well-being; AIQ = perceived AI-supported learning quality.

workflow as a scaffold for understanding; for pressure-aligned learners, it may be used more often as a shortcut or coping device. This interpretation is consistent with SDT's emphasis that autonomy support is associated with internalization and self-regulation, whereas controlling climates are more closely associated with controlled regulation and less adaptive functioning (Deci & Ryan, 2000; Ryan & Deci, 2000, 2020).

4.3. Contextual variation in the modeled associations

The multi-group difference tests indicated that the CM → COP association was significantly stronger in the high-pressure high-school background group, operationalized in this sample as students who attended high school in China, than in the comparison group (see Table 8). Within the present dataset, this pattern is consistent with the possibility that in more high-stakes and controlling educational climates, controlled motivation may be more closely associated with reliance-oriented GenAI behavior. One possible interpretation is that when performance-avoidance goals are more salient, GenAI may be used less as a learning partner and more as a coping-oriented tool. This interpretation is broadly consistent with recent cross-cultural work suggesting that students' perceptions and enactments of GenAI can vary across learning contexts and that culture and norms are linked to how AI is understood and used in education (Harrington et al., 2025). At the same time, the country labels in this study serve only as an operationalization within the present sample; the theoretical emphasis is on pressure-intensive versus autonomy-supportive climates more broadly, rather than on national categories themselves. Whether a similar pattern would be observed in Western higher-education settings or other institutional environments remains an open empirical question.

To connect this finding to adjacent evidence without conflating domains, it is also useful to note that problematic technology use has been linked to learning-related outcomes in broader digital behavior research. For example, Meng et al. (2025) found that mobile phone addiction among Chinese college students was associated with lower learning engagement, with academic self-efficacy and physical activity operating as mediating mechanisms. Although mobile phone addiction is not equivalent to GenAI use, this literature supports the more general point that unregulated reliance on digital tools can coincide with weaker engagement-related learning processes, which conceptually aligns with the risk pattern reflected in the pressure → controlled motivation → compulsive-use association pattern modeled here.

4.4. Interpreting perceived AI-supported learning quality (AIQ) and the strength of claims

In this study, perceived AI-supported learning quality (AIQ) refers to students' self-reported evaluations of the quality of AI-supported learning in the current semester, rather than to objective learning gains or externally verified academic performance. The AIQ scale captures learners' evaluations of whether GenAI use supports understanding, produces useable study materials, and enables effective learning with greater confidence after checking or verifying information. Accordingly, the present results should be interpreted as evidence about students' perceived learning-related value of GenAI use, not as evidence of causal or objectively verified improvements in achievement. This distinction is important because adjacent research on AI-mediated assessment has also shown that AI-generated educational judgments should not be treated as automatically interchangeable with validated human or performance-based measures. For example, Zacharis and Papadakis (2025) found that AI-human agreement in coursework assessment was substantially weaker than human inter-rater agreement, underscoring the need to distinguish perceived usefulness from objective evidence of learning quality. Within that scope, the model results suggest that perceived AI-supported learning quality is associated not merely with whether students use GenAI, but with how GenAI is incorporated into learning through the motivational and behavioral processes captured in the model. Adaptive AI use was positively associated with perceived academic performance and well-being, and both were positively associated with AIQ. Conversely, compulsive AI use was associated with lower well-being, which in turn was linked to lower AIQ. These patterns are consistent with the interpretation that GenAI may be experienced as more learning-supportive for some students and more reliance-oriented for others, while remaining within the limits of cross-sectional and self-reported data.

4.5. Implications for digital well-being, pedagogy, and educational data analytics

These findings have implications for human-centered design, educational practice, and educational data analytics.

Designing for autonomy, not just efficiency. Current GenAI interfaces often prioritize frictionless speed. In light of the association pattern linking pressure, controlled motivation, and compulsive AI use, designers and institutions may consider autonomy-supportive features that

introduce beneficial friction (e.g., prompts for intent clarification or verification checks) to reduce automaticity in vulnerable users.

Addressing the motivational gap. Universities and developers often focus on AI literacy (skills). However, the results of the dual-pathway model suggest that motivational quality is closely linked to whether those skills are enacted in more adaptive or more compulsive ways. Interventions that reduce environmental pressure and support student agency may be important, as AI anxiety and dependence are often discussed as symptoms of broader systemic stressors (Klimova & Pikhart, 2025).

Pedagogical and analytics implications. From an education perspective, the model points to teachable leverage points, especially autonomy support and planned trigger readiness. Instructors can scaffold planful GenAI workflows by requiring staged use (e.g., planning, drafting, verifying, revising) and by evaluating process and reasoning rather than only final answers. For example, instructors can ask students to pause before opening an AI tool and first write down three things: the purpose of use, the specific step of the task they need help with, and the criteria they will use to check the output. A simple classroom routine, such as outlining one's reasoning or prompt intention before consulting AI, may help cultivate more planful and less reliance-oriented use.

For educational data analytics, the framework suggests going beyond frequency counts toward patterns that reflect planful, iterative use versus rapid, minimally checked reliance. Where ethically permissible and transparent, analytics based on workflow traces (e.g., iteration depth, verification behaviors, revision histories) could be used to support learners, identify when additional scaffolding is needed, and evaluate whether autonomy-supportive course designs are associated with healthier engagement patterns. More broadly, the framing supports an educational innovation perspective that shifts GenAI guidance from tool training alone to motivation- and routine-aware pedagogy (e.g., autonomy-supportive course design plus explicit trigger-verification routines).

4.6. Limitations and future suggestions

Several limitations should be acknowledged when interpreting these findings. First, the cross-sectional survey design means that the directional paths specified in our model remain theoretically motivated rather than causally demonstrated; reciprocal or bidirectional relations cannot be ruled out. Second, all variables were assessed via self-report, including retrospective evaluations of high-school experiences and current AI-use patterns, which are susceptible to recall bias, social desirability, and shared method variance. Third, the key constructs of planned trigger readiness, adaptive AI use, and compulsive AI use were newly developed for the GenAI context; although their reliability and convergent validity were satisfactory, further work is needed to establish their discriminant validity vis-a-vis established measures of habit strength, problematic technology use, and digital well-being. Fourth, the sample was restricted to Chinese universities and Thai international colleges, with non-China-HS group school backgrounds largely concentrated in Thailand or international curricula. Accordingly, the present findings are best interpreted as evidence from China, Thailand-linked educational settings rather than as confirmation of a universally generalizable GenAI pathway structure. To reduce identifiability risk and keep the questionnaire concise, we did not collect sensitive demographics (e.g., gender, age), and thus cannot fully rule out demographic confounding; future studies should include these variables to test robustness and the modeled relations within the dual-pathway framework.

At the same time, these Western and English-speaking studies should not be treated as direct replications of the present model. First, the same construct assumption is not fully met. Much of the existing literature focuses on student perceptions, acceptable uses, academic-integrity concerns, or lived experience, whereas the present study models retrospective high-school motivational climate, autonomous versus

controlled motivation, planned trigger readiness, and adaptive versus compulsive AI-use patterns. Second, the same measurement assumption is also not met. Prior studies have often used descriptive surveys, qualitative interviews, or adoption-oriented frameworks rather than the SDT-based latent-variable structure developed here. Third, the same mechanism assumption therefore remains open. Cross-regional evidence from the Nordic countries and the USA suggests that some broad drivers of student ChatGPT adoption recur across settings, but the relative importance of specific predictors still differs by region (Faraon et al., 2025). The present findings should therefore be interpreted not as confirmation of a universally invariant psychological mechanism, but as a theory-informed mechanism hypothesis supported within China, Thailand-linked contexts.

These limitations also suggest several productive directions for future research. Longitudinal designs that follow students from late high school into university would allow stronger tests of temporal ordering and could clarify whether these adaptive and compulsive patterns reflect relatively stable tendencies or more fluid states over time. Experimental and quasi-experimental work could manipulate autonomy-supportive teaching practices, AI-use guidelines, or interface “friction” to test whether strengthening Current University Autonomy Support or scripting “healthy” AI triggers can attenuate reliance-oriented patterns and reinforce adaptive engagement. Future studies could also combine surveys with behavioral and learning analytics data (e.g., AI log traces, assignment histories, or experience sampling) to capture the micro-dynamics of trigger-action cycles more directly and extend the present comparisons to additional regions, disciplines, and institutional types. Finally, qualitative or mixed-methods work with students and instructors could deepen understanding of how GenAI is experienced as either an “escape valve” or a “learning partner,” providing richer grounding for policy, training, and design efforts aimed at supporting both learning outcomes and digital well-being.

4.7. Conclusion

This study proposed and tested a dual-pathway model of GenAI use that integrates Self-Determination Theory with habit-formation perspectives to examine how distal educational climates and current university contexts are associated with students' digital behaviors and perceived outcomes. Addressing RQ1 to RQ3, the results were consistent with the idea that high-school pressure and autonomy support, together with Current University Autonomy Support, are associated with distinct motivational profiles and patterns consistent with either an adaptive pattern, characterized by autonomous motivation, planned trigger readiness, adaptive AI use, and higher perceived performance, well-being, and perceived AI-supported learning quality, or a compulsive pattern, in which pressure-aligned controlled motivation is associated with more compulsive AI use and lower well-being. These interpretations are supported by the reported bootstrapped indirect effects in the full sample and subgroup analyses (Table 4; Tables 9 and 12).

Through the multi-group analyses (RQ4 and RQ5), the overall pattern of associations appeared broadly similar across the sampled high-school and university groupings, while one key link between controlled motivation and compulsive AI use was stronger in the high-pressure high-school background group as operationalized in this sample (Table 8). Taken together, the findings suggest that the central challenge for GenAI in education is not simply whether students use AI, but how motivational quality and habit-like routines are associated with whether AI functions more as a scaffold for learning or as a coping-oriented shortcut. By foregrounding autonomy support, planned trigger readiness, and context-sensitive pressure, the model offers a useful framework for future research and for the design of autonomy-supportive pedagogies and AI systems that encourage more adaptive, rather than more compulsive, engagement with GenAI in higher education. However, these interpretations should be understood within the China, Thailand-linked contexts examined here, and further

comparative research is needed before extending them to other educational systems.

CRediT authorship contribution statement

Shen Qinjie: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wynn Arunrugstichai:** Writing – original draft, Supervision, Project administration.

Ethics statement

This study involved human participants and received an ethics exemption determination from the institutional ethics committee at Raffles International College Bangkok (Ethics Exemption ID: RICB-ETH-2025-019). Participation was voluntary and anonymous. Before starting the online questionnaire, participants read an information sheet and provided informed consent electronically. To protect privacy and confidentiality, the survey did not collect direct identifiers (e.g., names, student IDs, or contact details). Data are reported only in aggregate and, where applicable, in de-identified form consistent with the exemption conditions.

Data availability statement (open data)

The data that support the findings of this study are available from the corresponding author upon reasonable request. Due to ethical and

privacy considerations (including the protection of participant confidentiality and the conditions of the ethics exemption), the dataset is not publicly posted. Any shared data will be provided in a de-identified form and may be subject to reasonable access conditions consistent with the approved protocol and applicable ethical requirements.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used ChatGPT for language polishing and draft refinement. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.caeai.2026.100621>.

Appendix A. English Version of Survey Instrument

General Instruction

Unless otherwise noted, please answer based on this semester.

7-point scale

1 = Strongly disagree; 2 = Disagree; 3 = Slightly disagree; 4 = Neutral; 5 = Slightly agree; 6 = Agree; 7 = Strongly agree.

Block 1. Consent

You will complete an anonymous academic survey about learning with generative AI. We do not collect school names or student IDs. Your data will be analyzed in aggregate only. Participation is voluntary, and you may exit at any time. Estimated time: 15 min. Click “Start” to consent.

Response: I consent.

Block 2. Screening & Demographics

We will group respondents by where they completed the majority of Grades 10–12 (China vs Thailand).

HS COUNTRY - High-School Country (Grades 10–12 majority)

Where did you complete the majority of Grades 10–12?

Options: China (mainland); Thailand; Other.

HS CURRICULUM - High-School Curriculum

Which curriculum best describes your high-school program?

Options: China National (Gaokao); Thai National; International (IB/AP/A-Level/Other); Other.

HS LANGUAGE - High-School Language of Instruction

Main language of instruction in high school:

Options: Chinese; Thai; English; Other.

COUNTRY UNI - Current University Country

Where are you currently studying (university location)?

Options: China; Thailand; Other.

CONTEXT UNI - Current Study Context

Which study context best describes you this semester?

Options: Chinese university; Thai international university/college; Other.

RESPONSE LANG

What is the language you used to answer this survey?

Options: English; Chinese.

AI USE FREQ

This semester, how often do you use generative AI for learning per week?

Options: Never; <1 time; 1–2 times; 3–5 times; 6–10 times; >10 times.

AI EXPERIENCE MO

For how long have you been using generative AI tools?

Options: <3 months; 3–6 months; 7–12 months; 13–24 months; >24 months.

MAJOR - Major Area (Broad Category)

Your major area (broad category).

Options: Engineering; Science; Arts & Humanities; Social Sciences; Business & Management; Art/Design/Media; Medicine/Health; Other.

YEAR - Study Year/Level

Your current study year/level.

Options: 1st year; 2nd year; 3rd year; 4th year; Postgraduate; Other.

BOARDING - Living on Campus

Do you live on campus (boarding)?

Options: Yes; No; Prefer not to say.

STUDY HRS - Weekly Study Hours (Outside Class)

Approximate weekly study hours (outside class).

Options: <5 h; 5–10 h; 10–15 h; 15–20 h; 20–30 h; >30 h.

*Block 3. High-School Experiences**High-School Pressure (HSP)*

HSP1 In my high school, grades/ranks/scholarship competition were frequently emphasized.

HSP2 I often felt I had to meet performance standards to avoid penalties or negative judgments.

HSP3 Peer comparison and ranking were common.

HSP4 Teachers/school placed strong emphasis on exam scores.

High-School Autonomy Support (HSAS)

HSAS1 Teachers allowed choices in assignment topics or formats.

HSAS2 I could progress at a self-paced speed.

HSAS3 Materials of different interests/difficulties were available.

HSAS4 Teachers encouraged multiple ways to demonstrate understanding.

*Block 4. Current University Context (This Semester)**Current University Autonomy Support (CAS)*

- CAS1 My courses/instructors provide choices in tasks or learning paths this semester.
- CAS2 I am able to adjust pacing or deadline windows when needed this semester.
- CAS3 The course/system (including AI tools) offers options with different difficulty or explanation depth this semester.
- CAS4 I can use different output formats (e.g., report, presentation, visualization) this semester.

*Block 5. Traits (Motivation & Trigger Readiness) – This Semester**Controlled Motivation (CM)*

- CM1 I study mainly to avoid criticism or disappointing others this semester.
- CM2 I would feel guilty or anxious if I didn't study hard this semester.
- CM3 I often feel compelled by external demands or evaluations this semester.
- CM4 I worry about problems if I don't meet expectations this semester.

Autonomous Motivation (AM)

- AM1 I study because I am genuinely interested in the material this semester.
- AM2 Studying feels meaningful and valuable to me this semester.
- AM3 I identify with the importance of my learning goals this semester.
- AM4 I enjoy understanding new concepts this semester.

Planned Trigger Readiness (PTR)

- PTR1 At set times/places, I generally start studying/using AI this semester.
- PTR2 I keep low-friction starters (e.g., checklists/templates/prompt library) this semester.
- PTR3 I usually plan steps before I start this semester.
- PTR4 I can maintain my study rhythm when following my plan this semester.

*Block 6. AI-Use Pathways**Compulsive AI-Use Pathway (COP)*

- COP1 This semester, I tend to paste questions into AI to get a ready-to-use answer as quickly as possible.
- COP2 This semester, when AI gives a plausible answer, I often move on with minimal checking.
- COP3 This semester, I rely on AI to produce final wording with only minor edits.
- COP4 This semester, I tend to extend AI use mainly to push for a quick result.

Adaptive AI-Use Pathway (ADP)

- ADP1 This semester, I use AI to plan steps and decide what I should do next.
- ADP2 This semester, I ask AI for explanations until I reasonably understand.
- ADP3 This semester, I rework AI outputs into my own notes (with sources or examples when needed).
- ADP4 This semester, I check or compare AI results before using them in my work.

*Block 7. Behavioral Functioning (This Semester)**Perceived Academic Performance (APerf)*

- APerf1 This semester, I usually complete my coursework on time.
- APerf2 This semester, I keep up with most academic requirements in my classes.
- APerf3 This semester, the quality of my academic work generally meets instructors' expectations.
- APerf4 This semester, I manage my studies effectively overall.

Subjective Well-being (WB)

- WB1 This semester, I generally feel positive and energetic.
- WB2 This semester, I maintain a workable balance between study and life.
- WB3 This semester, I get adequate sleep/rest for my needs.
- WB4 This semester, I can manage study-related stress in a workable way.

*Block 8. Final Outcome (This Semester)**Perceived AI-supported learning quality (AIQ)*

- AIQ1 This semester, using AI helps me deepen my understanding.
- AIQ2 This semester, AI outputs are useable and can be organized into reliable study materials.
- AIQ3 This semester, after checking, I am generally confident in AI outputs' accuracy/consistency.

AIQ4 This semester, AI helps me work more efficiently and with more assurance.

Block 9. Quality Control & Debrief

AC1 (Attention Check)

To confirm attention, please select “Agree”.

Options: Strongly disagree; Disagree; Slightly disagree; Neutral; Slightly agree; Agree; Strongly agree.

AC2 (Attention Check)

For quality check, please select the number “5”.

Options: 1; 2; 3; 4; 5; 6; 7.

OPEN1 (optional)

In 1–2 sentences, briefly describe how you typically use AI for learning.

Response: Open text.

Appendix Table A1

Summary of confirmatory factor analysis: Factor loadings, composite reliability, and average variance extracted (full sample)

Construct	Number of items	Loading range	CR	AVE
HSP	4	0.955-0.960	0.978	0.918
HSAS	4	0.965-0.967	0.982	0.933
CAS	4	0.951-0.956	0.976	0.910
CM	4	0.918-0.932	0.959	0.853
AM	4	0.933-0.939	0.966	0.876
PTR	4	0.940-0.945	0.969	0.888
COP	4	0.890-0.911	0.945	0.811
ADP	4	0.934-0.944	0.968	0.884
APerf	4	0.928-0.945	0.967	0.878
WB	4	0.928-0.939	0.965	0.872
AIQ	4	0.942-0.946	0.970	0.890

Note. Convergent validity was supported because AVE values exceeded the recommended 0.50 threshold. Discriminant validity was supported based on AVE-based criteria (i.e., AVE exceeded squared inter-construct correlations; square roots of AVE were larger than inter-factor correlations).

Appendix Table A2

Measurement invariance tests across grouping schemes

Panel	Model	Constraint type	CMIN/df	CFI	RMSEA	Delta CFI	Delta RMSEA
A (High-school background: China-HS group vs non-China-HS group)	Model 1	Configural	1.84	0.981	0.035	-	-
A (High-school background: China-HS group vs non-China-HS group)	Model 2	Metric	1.85	0.979	0.036	0.002	0.001
A (High-school background: China-HS group vs non-China-HS group)	Model 3	Scalar	1.89	0.975	0.037	0.004	0.001
B (University location: China-Uni group vs Thailand-Uni group)	Model 1	Configural	1.98	0.978	0.036	-	-
B (University location: China-Uni group vs Thailand-Uni group)	Model 2	Metric	2.01	0.975	0.037	0.003	0.001
B (University location: China-Uni group vs Thailand-Uni group)	Model 3	Scalar	2.06	0.971	0.038	0.004	0.001

Note. Configural = same factor structure across groups with parameters freely estimated; Metric = factor loadings constrained equal across groups; Scalar = factor loadings and intercepts constrained equal across groups. Delta CFI and Delta RMSEA indicate changes relative to the less constrained model.

Appendix B. Chinese Version of Survey Instrument

统一作答说明

除非另有说明,以下题目均以本学期为参照作答。

7点量表

1 = 非常不同意; 2 = 不同意; 3 = 稍不同意; 4 = 一般; 5 = 稍同意; 6 = 同意; 7 = 非常同意。

第1部分 同意说明

本研究是一项关于使用生成式AI进行学习的匿名学术问卷。我们不收集学校名称或学号,数据仅以整体统计呈现。参与自愿,可随时退出。预计耗时15分钟。点击“开始”即表示同意参与。

作答:我同意。

第2部分 筛选与基本信息

分组原则:我们将主要根据你高中10–12年级多数所在国家进行分组(China vs Thailand)。

HS COUNTRY – 高中所在国家(10–12年级多数)

你的高中10–12年级多数年级主要在哪个国家就读?

选项: 中国大陆; 泰国; 其他.

HS CURRICULUM – 高中课程体系

你的高中课程体系是以下哪一种?

选项: 中国高考体系; 泰国本土体系; 国际课程(IB/AP/A-Level/其他); 其他.

HS LANGUAGE – 高中授课语言

高中主要授课语言:

选项: 中文; 泰语; 英语; 其他.

COUNTRY UNI – 当前大学所在国家

你目前就读的大学所在国家/地区是?

选项: 中国; 泰国; 其他.

CONTEXT UNI – 当前学习情境

本学期你的学习情境最符合:

选项: 中国高校; 泰国国际学院; 其他.

RESPONSE LANG – 作答语言

你本次作答主要使用的语言是?

选项: 英文; 中文.

AI USE FREQ – AI 使用频率

本学期,你每周用于学习的生成式AI使用频率是?

选项: 从不; 少于1次; 1–2次; 3–5次; 6–10次; 多于10次.

AIE EXPERIENCE MO – 使用时长

你使用生成式AI工具的累计时间是多久?

选项: 少于3个月; 3–6个月; 7–12个月; 13–24个月; 超过24个月.

MAJOR – 专业大类

你的专业大类.

选项: 工程; 理学; 人文与艺术; 社会科学; 商业与管理; 设计与传媒; 医学与健康; 其他.

YEAR – 年级/学习阶段

你的年级/学习阶段.

选项: 大一; 大二; 大三; 大四; 研究生; 其他.

BOARDING – 是否住校

你是否住校(宿舍)?

选项: 是; 否; 不愿透露.

STUDY HRS – 每周课外学习时长

每周课外学习时长(大致).

选项: 少于5小时; 5–10小时; 10–15小时; 15–20小时; 20–30小时; 多于30小时.

第3部分 高中经历(回溯)

注:本模块为高中回溯,非“本学期”。

高中绩效压力(HSP)

- HSP1 在我的高中,成绩/排名/奖学金竞争经常被强调。
- HSP2 我常常觉得必须达到既定标准,否则会受到惩罚或负面评价。
- HSP3 同学之间的比较与排名很常见。
- HSP4 老师/学校对考试分数非常重视。

高中教师自主支持(HSAS)

- HSAS1 老师允许在作业主题或呈现方式上做出选择。
- HSAS2 我可以按照适合自己的节奏推进学习。
- HSAS3 我可以选择不同兴趣或难度的学习材料。
- HSAS4 老师鼓励用多种方式展示理解。

第4部分 当前大学情境(本学期)

当前自主支持(CAS)

- CAS1 本学期,我的课程/老师提供任务或学习路径的选择。
- CAS2 本学期,在需要时我可以调整学习节奏或截止窗口。
- CAS3 本学期,课程/系统(含AI工具)提供不同难度或解释深度的可选项。
- CAS4 本学期,我可以用不同呈现形式完成作业(如报告/演示/可视化等)。

第5部分 特质层(动机与触发准备度,本学期)

受控动机(CM)

- CM1 本学期,我主要为了避免被批评或让别人失望而学习。
- CM2 本学期,如果我不努力学习,会感到内疚或焦虑。
- CM3 本学期,我常因外在要求或评价而被迫学习。
- CM4 本学期,如果达不到期望,我会担心会有麻烦。

自主动机(AM)

- AM1 本学期,我学习是因为对内容真正感兴趣。
- AM2 本学期,学习对我而言是有意且有价值的。
- AM3 本学期,我认同自己的学习目标的重要性。
- AM4 本学期,我享受理解新概念的过程。

计划性触发准备度(PTR)

- PTR1 本学期,到固定时间或地点时,我通常会开始学习/使用AI。
- PTR2 本学期,我保留低摩擦启动方式(如清单/模板/提示词库)。
- PTR3 本学期,我通常会在开始前规划步骤。
- PTR4 本学期,当我按照计划执行时,能够保持学习节律。

第6部分 AI 使用路径(本学期)

强迫性过度 AI 使用路径(COP)

- COP1 本学期,我倾向于把问题直接粘贴给AI,以尽快获得可直接使用的答案。
- COP2 本学期,当AI给出看起来可行的答案时,我常只做最少核查就继续。
- COP3 本学期,我依赖AI生成接近最终用语,仅做少量修改。
- COP4 本学期,我倾向于延长AI使用,主要是为了更快得到结果。

健康习惯 AI 使用路径(ADP)

- ADP1 本学期,我使用AI来规划步骤并决定下一步要做什么。
- ADP2 本学期,我会向AI请求解释,直到我基本理解为止。
- ADP3 本学期,必要时我会把AI输出重新整理为自己的笔记(并在需要时加入来源或例子)。
- ADP4 本学期,我在把AI结果用于作业前会进行核对或比对。

第7部分 功能层(本学期)

学业表现(自我感知,APerf)

- APerf1 本学期,我通常能够按时完成课程学习任务。
- APerf2 本学期,我能跟上大多数课程的学习要求。
- APerf3 本学期,我的学习成果整体上符合任课老师的期望。

APerf4 本学期,从整体来看,我能够较为有效地管理学习。
 主观幸福感(WB)
 WB1 本学期,我整体感觉积极、有活力。
 WB2 本学期,我在学习与生活之间保持一种可行的平衡。
 WB3 本学期,我获得了与自身需要相匹配的睡眠/休息。
 WB4 本学期,我能够以一种可行的方式管理与学习相关的压力。

第8部分 最终因变量(本学期)

AI 学习质量知觉(AIQ)

AIQ1 本学期,使用AI有助于加深我的理解。
 AIQ2 本学期,AI输出可用,且可整理为可靠的学习资料。
 AIQ3 本学期,在核查后,我总体上对AI输出的正确性/一致性有信心。
 AIQ4 本学期,AI有助于我更高效且更有把握地完成任务。

第9部分 质控与去敏

AC1 注意力检验

为确认你认真作答,请选择“同意”。
 选项: 非常不同意; 不同意; 稍不同意; 一般; 稍同意; 同意; 非常同意。

AC2 注意力检验

为质检,请选择数字“5”。
 选项: 1; 2; 3; 4; 5; 6; 7。

OPEN1(可选)

请用1–2句话简要描述你通常如何使用AI进行学习。
 作答方式: 开放式文本。

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